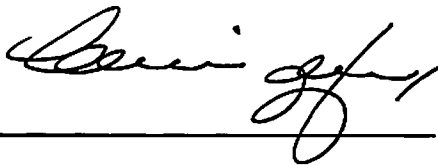


EXHIBIT A

UNITED STATES V. CEMEX, INC.
Civil Action No. 09-cv-00019-MSK-MEH
U.S. District Court for the District of Colorado

Expert Report

Phyllis Fox, Ph.D., PE



On Behalf of the United States of America

March 1, 2011

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I. BACKGROUND

A. Information Required by the Federal Rules of Civil Procedure

The following is a list of items required by the Federal Rules of Civil Procedure:

- This report contains my opinions, conclusions, and reasons therefore.
- Appendix A contains a statement of my qualifications and publications.
- Appendix B lists the reports, data, and other information I considered in forming my opinions.
- Appendix C contains a list of my previous testimony as an expert witness at trial or by deposition in the last four years.
- I am being compensated at the rate of \$200/hr for my work on this matter.

The opinions expressed in this report are my own and are based on the data and facts available to me at the time of writing. Should additional relevant or pertinent information become available, I reserve the right to supplement the discussion and findings in my report.

B. Purpose of Report

I was asked by the United States Department of Justice to review changes at the CEMEX, Inc., (CEMEX) Lyons Plant made between 1997 and 2000, to determine if Prevention of Significant Deterioration (PSD) review would have been required for nitrogen oxides (NO_x) prior to their construction. I was asked to analyze these changes in two ways: (1) collectively, as a single project (Case A); and (2) individually (Case B). I was also asked to evaluate a range of production goals and baseline assumptions to bracket the net increase in NO_x emissions.

C. Summary of Experience

I am an independent environmental engineer based in Rockledge, Florida. My resume is attached at Appendix A.

I received my Ph.D. in environmental/civil engineering from the University of California at Berkeley in 1980. I am a Registered Professional Engineer in five states and Board certified in Air Pollution Control by the American Academy of Environmental Engineers. I am listed in *Who's Who in Science and Engineering*, *Who's Who in the World*, and *Who's Who of American Women*.

I have worked as a consulting environmental engineer since 1981. Previously, I served as Principal Investigator at Lawrence Berkeley National Laboratory ("LBNL") for four years and as an Engineer at Bechtel, Inc., for five years. At LBNL, I directed a large research program on environmental impacts and control technologies for the synthetic

fuels industry. At Bechtel, I worked on the design and assessment of power plants and refineries.

I have 40 years of experience in applying a wide range of regulations to environmental impacts, permitting, and reporting for power plants, refineries, steel mills, cement plants, and many other types of industrial facilities. This experience includes: air permitting; emission calculations; control technology determinations, including best available control technology (“BACT”), best available retrofit technology (“BART”), lowest achievable emissions rate (“LAER”), and maximum achievable control technology (“MACT”); emission offsets; enforceability of air permits; and rulemakings. I have performed emission calculations and control technology analyses for more than 300 industrial facilities.

II. SUMMARY OF CONCLUSIONS

I have assessed the applicability of PSD requirements to a collection of physical and operational changes made at CEMEX’s Lyons Plant between 1997 and 2000. I evaluated these changes both collectively and individually. My analyses indicate that these changes result in a significant net increase in NO_x emissions when evaluated both collectively and individually. Thus, these changes are major modifications and would have triggered PSD review prior to construction. This would have required CEMEX to obtain a PSD permit prior to the start of construction and to apply best available control technology (BACT). The following sections explain the regulatory framework that applies to a PSD analysis and present the relevant emission calculations that support my conclusions.

III. THE 1997 – 2000 CHANGES

To increase its capacity to produce cement, CEMEX made a series of changes to the Lyons Plant between 1997 and 2000. These changes are described in Walter Greer’s Expert Report and are summarized here in Table 1.¹ Producing more cement required increasing the throughput of the pyroprocessing unit, upstream operations (quarry, raw mill), and downstream operations (finish mill).²

¹ In Table 1, for each change, I have also identified the date construction started on that change.

² Expert Report of Walter L. Greer, p. 26 – 40.

**Table 1:
Changes Made at the Lyons Plant
1997 – 2000**

	Project Name	Start Construction	Type of Change	Bates Number
1	Combustion Chamber Modifications	3/97	Pyro	CMXLYO00049753 CMXLYO00180115
2	Raw Mill Transfer Line	5/97	Debottle	CMXLYO00089627
3	Purchased Oxygen Injection	5/97	Pyro	CMXLYO00051582 CMXLYO00049554
4	ID Fan 1500 HP Motor	8/97	Pyro	CMXLYO00453897 CMXLYO00180115
5	New Raw Feed Compressor	12/97	Debottle	CMXLYO00052369
6	New High Efficiency Separator	10/98	Debottle	CMXLYO00014980 CMXLYO00015120
7	Quarry Expansion	10/98	Debottle	CEMEX_NEIC_00000257 ³
8	Tempering Tee Damper	1/99	Debottle	CEMEX_NEIC_00003484 CEMEX_NEIC_00003493
9	VSA Oxygen Plant	5/99	Pyro	CEMEX_NEIC_00003499 CMXLYO00045641
10	New Annulus Boost Fan	8/00	Pyro	CMXLYO00157315 CMXLYO00045597

Pyro = pyroprocessing unit; **Debottle** = debottlenecking.

Between 1997 and 2000, two types of changes were made at the Lyons Plant: (1) changes to the pyroprocessing unit that resulted in a direct increase in clinker production and NOx emissions and (2) changes to upstream and downstream material handling processes that indirectly increased NOx by debottlenecking the pyroprocessing unit. In Table 1, the changes that resulted in direct increases in clinker production are numbers #1, #3, #4, #9, and #10. The balance of the changes listed in Table 1, #2 and #5 – #8, increased the throughput of the quarry, raw mill, and finish mill so the changes to the pyroprocessing unit could be realized. The maximum increase in clinker production from the pyroprocessing unit could not be realized without changes to these upstream and downstream processing units.⁴ Changes #2 and #5 – #8 are classic debottlenecking changes. The increase in NOx emissions from these debottlenecking changes must be

³ I have assumed construction started immediately after issuance of modified quarry permit 98-BO-1414-F, September 11, 1998, which is consistent with the new high efficiency separator construction start date and the push to complete modifications prior to end of year, due to impending permit violations. See, Lyons Plant—Monthly Report of Operations, January 1998, CMXLYO00050091 at 50143 (“It is essential that this work be completed as early as possible to avoid possible NOV’s for exceeding permitted tonnage’s (sic) before the end of 1998.”).

⁴ Expert Report of Walter L. Greer, p. 32.

included in the emission calculations to determine PSD applicability.⁵ Although some of the debottlenecking changes were issued minor source permits in 1998 for their particulate matter emissions (PM₁₀),⁶ none of these permitting actions considered the increase in NO_x emissions from changes at upstream and downstream processing units that debottlenecked the pyroprocessing unit or the changes at the pyroprocessing unit itself.

IV. MAJOR MODIFICATION APPLICABILITY

The Lyons Plant is a major stationary source that is located in an area that is designated as in attainment with national ambient air quality standards for NO_x.⁷ A major modification is “any physical change in or change in the method of operation of a major stationary source that would result in: a significant emissions increase... of a regulated NSR pollutant...; and a significant net emissions increase of that pollutant from the major stationary source.”⁸

I was asked to determine if the changes in Table 1, taken both collectively (Case A) and individually (Case B), are a major modification for NO_x. As the Lyons Plant is located in an attainment area, I determined if PSD review was required. This determination involves a three-step process: (1) confirming that changes are physical changes or changes in the method of operation that are not exempt; (2) determining emission increases from the subject change(s) and if Step 2 emissions are significant; (3) determining the net change in emissions.⁹

Step 1 evaluates whether the subject changes are physical changes or changes in the method of operation that are not otherwise exempt. As explained in Mr. Greer’s Expert Report, the subject changes are physical changes (addition/modification of injection points, separators, fans, motors, transfer lines, compressors, calciner geometry,

⁵ EPA, New Source Review Workshop Manual, Prevention of Significant Deterioration and Nonattainment Area Permitting, Draft, October 1990 (hereafter “NSR Manual”), www.epa.gov/ttn/nsr/gen/wkshpman.pdf, p. A.46 (“[Step 1] First, only the emissions increases expected to result from the proposed project are examined. This includes emissions increases from the new and modified emissions units and any other plant-wide increases (*e.g.*, debottlenecking increases) that will occur as a result of the proposed modification.”).

⁶ Southdown, Inc., Construction Permit and Permit Modification Application, April 30, 1998, CMXLYO00015334 at 15582 (Overview of all permits); CDPHE – APCD, Construction Permits: Permit No. 98-BO-0259, September 11, 1998, CEMEX_NEIC_00000246 (New finish mill); Permit No. 98-BO-0292, September 11, 1998, CEMEX_NEIC_00000229 (New raw materials storage and handling at the plant site); Permit No. 93-BO-1414-F, September 11, 1998, CEMEX_NEIC_00000257 (Modifications to Dowe Flats Quarry).

⁷ Defendant CEMEX, Inc., Amended First Set of Responses to United States’ First Set of Requests for Admission, Response to Request for Admission No. 8.

⁸ 40 CFR 52.21(b)(2)(i); CAQCC Reg. 3, I.B.2 (1986); CAQCC Reg. 3, Part A, I.B.35.B (1997). *See*, NSR Manual Excerpt, p. A.33 – A.34, US00035301, which explains how this has been interpreted by EPA. *See also*, Deposition of Bradley Cooley, September 20, 2010, TR at 27:8 – 12, 27:21 – 28:8.

⁹ *See* discussion in NSR Manual, p. A.33 – A.56.

etc.) or changes in the method of operation (oxygen injection, debottlenecking, increases in throughput limits, etc.) that increased NOx emissions.¹⁰

If a subject change is a physical change or change in the method of operation and no exemptions apply, Step 2 determines whether the change(s), taken collectively (Case A) or individually (Case B) would result in a significant NOx emission increase. This is determined by calculating the difference between a modified emission unit's "actual emissions"¹¹ during a baseline period prior to the change (the "baseline period") and the unit's potential to emit after the change.¹²

If the resulting emission increase in Step 2 equals or exceeds the PSD significance threshold, Step 3 calculates the net emission increase, which considers creditable and contemporaneous increases and decreases from the entire facility for the period commencing 5 years before the expected date of construction and ending at the time when the emission increase occurs.¹³ The emissions increase resulting from a change occurs when the new emission unit becomes operational and begins to emit.¹⁴ To be creditable, however, any decrease must be enforceable prior to the commencement of actual construction.¹⁵

I assessed the PSD implications of two major groupings of these changes. In Case A (Figure 1), I assume that all of the changes identified in Table 1 were part of a single project, the goal of which was to increase cement production at the Lyons Plant. In Case B (Figure 2), I evaluate each of the ten changes individually.

For Case A, I evaluate two separate clinker production goals (546,250 ton/yr and 577,920 ton/yr),¹⁶ and for each clinker production goal, I evaluate two separate baseline periods: (1) 24 months prior to the start of construction (Scenarios "1" and "3"); and (2)

¹⁰ Expert Report of Walter L. Greer, p. 26 – 40.

¹¹ "Actual emissions" is a term of art for purposes of the Clean Air Act and PSD regulations. *See*, 40 CFR § 52.21(b)(48).

¹² *See, e.g.*, NSR Manual, p. A.33 – A.34, A.45 ("Determine the emissions increases (but not any decreases) from the proposed project. If increases are significant, proceed; if not, the source is not subject to review."), A.46 ("Step 1. First, only the emissions increases expected to result from the proposed project are examined. This includes emission increases from the new and modified emissions units and any other plant-wide emissions increases (*e.g.*, debottlenecking increases) that will occur as a result of the proposed modification."); Deposition of Keith Baugues, July 30, 2010, TR at 48:1 – 5.

¹³ 40 CFR 52.21(b)(3)(i). NSR Manual, p. A.37 – A.42.

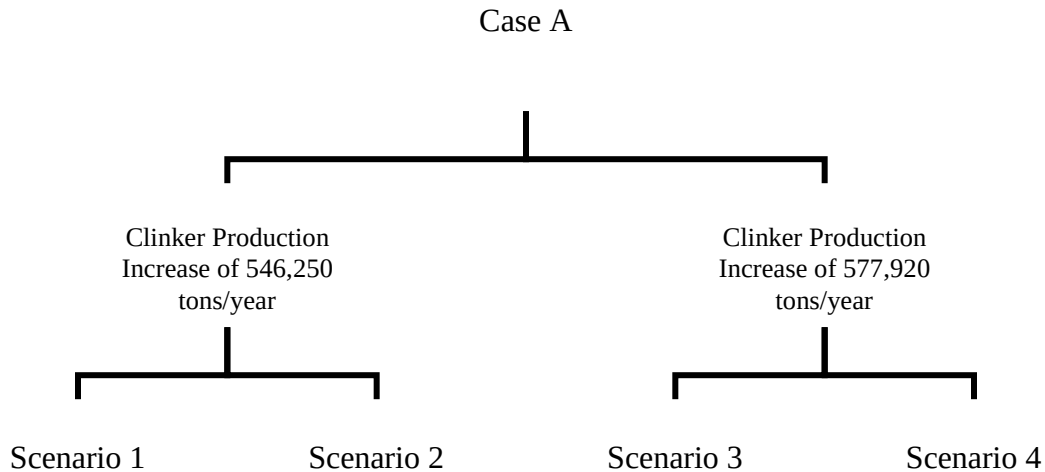
¹⁴ NSR Manual, p. A.38 ("An increase resulting from a physical change at a source occurs when the new emissions unit becomes operational and begins to emit a pollutant.").

¹⁵ *See*, 40 CFR 52.21(b)(3)(vi). *See also*, NSR Manual, p. A.38 ("To be creditable, a contemporaneous reduction must be federally enforceable on and after the date construction on the proposed modification begins.").

¹⁶ A range of production goals was selected based on CEMEX's targets, noted in its various Approvals for Expenditure and Capital Budgets. *See, infra*, Section IV.B.1.b.i: Maximum Clinker Production for the Collective Changes (Case A, Step 2).

standing in the shoes of a permit engineer (Scenarios “2” and “4”).¹⁷ Thus, four separate scenarios are analyzed in Case A.

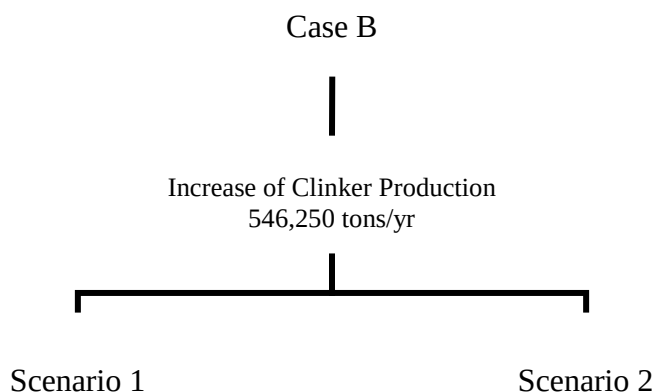
**Figure 1:
Scenarios Analyzed for PSD Applicability (Collective Changes)**



For Case B, I evaluated the increase in clinker production for each change identified in Table 1, based on the lower end of CEMEX’s estimate of the impact that change had on clinker production levels. In Case B, I used the same two baseline periods as for Case A but only made a single calculation for the lower end of the production range. Thus, in Case B, two separate scenarios are analyzed for each change to bracket the increase in NOx emissions, corresponding to different baseline periods.

¹⁷ The term “standing the shoes of a permit engineer” refers to the historic time period and information that would have then been available to a permit engineer at the time that each change should have been permitted.

**Figure 2:
Scenarios Analyzed for PSD Applicability (Individual Changes)**



A. Exemptions (Step 1, Cases A & B)

The applicable regulations do not define “physical change” or “change in the method of operation.” The regulations, however, do specifically exclude from these activities certain types of events, unless such change would be prohibited under any federally enforceable permit conditions established after January 6, 1975.¹⁸ Based on my review of the produced documents and Mr. Greer’s Expert Report,¹⁹ I conclude that all of the changes listed in Table 1 should be included in my emission calculations, as they are all non-exempt physical changes or changes in the method of operation.²⁰ Thus, the following sections discuss Steps 2 and 3 of the PSD determination for two cases: (1) Case A—all of the changes taken collectively as a single project; and (2) Case B—each change considered individually.

B. Case A: Collective Changes

In Case A, I evaluate the collection of changes in Table 1 as a single project. In my opinion, Case A is the correct way to determine PSD applicability because the changes are interrelated, they have been planned together, and their collective purpose is to increase source-wide production capacity.²¹

The nexus between the debottlenecking changes and pyroprocessing changes set forth in Table 1 is evident in the approved Authorization for Expenditure (AFE) for the

¹⁸ 40 CFR 52.21(b)(2)(iii)(f); CAQCC Reg. 3, I.B.2.b and c (1986); CAQCC Reg. 3, Part A, I.35.B.b and c (1997); NSR Manual, p. A.33 – A.34.

¹⁹ Expert Report of Walter L. Greer, p. 5 – 8.

²⁰ Expert Report of Walter L. Greer, p. 28 – 40.

²¹ Expert Report of Walter L. Greer, p. 7, 32.

“Lyons Plant Expansion.”²² For example, the AFE explicitly recognizes that “[a] portion of this capacity increase [125,000 ton/yr of cement] was achieved in 1997 through minor modifications to the preheater aerodynamics [Change #1] combined with the temporary use of high cost purchased oxygen [Change #3].”²³ This AFE goes on to explain that the production rates of the upstream changes (raw mill transfer line, raw feed compressor, tempering tee damper) and downstream changes (separator) must be increased to expand the plant capacity—specifically naming each change in Table 1 except the quarry.²⁴ The AFE even included the booster fan, the last change in the series, which was not implemented until August 2000, nearly two years after the AFE was approved. Thus, CEMEX’s own documents describe how the changes in Table 1 are interdependent.²⁵

1. Significant Emission Increase for Collective Changes (Case A, Step 2)

The second step is determining the difference between “actual emissions” during a baseline period prior to a change and the potential to emit after the change.²⁶ I calculated the increase in emissions due to the subject changes by subtracting the baseline actual NOx emissions preceding the start of construction of the first change from the future potential to emit NOx. I identified the modification of the calciner combustion chamber in March 1997 as the start of construction of the first change. The methods used to make these calculations are explained in the NSR Manual, which is a document that CEMEX, its permitting consultants, and the State relied upon.²⁷

²² Memorandum from Karen Krenek to Norris Widener, Re: Approved AFE, November 18, 1998, CEMEX_NEIC_00003449.

²³ Here and elsewhere in this report, the numbered changes [Change #] refer to the changes identified in Table 1 and described in the Expert Report of Walter L. Greer, p. 26 – 40.

²⁴ The produced documents do not include an AFE for the quarry work, but the record is clear that the work was required to debottleneck the kiln. *See, e.g.*, Lyons Plant – Monthly Report of Operations, January 1998, CMXLYO00050091 at 50143 (“This work [removing restrictions on Dowe Flats production] is extremely important and should allow for significantly higher mining levels that would match the higher throughputs on the kiln (results from the combustion chamber and oxygen injection in 1997).”); Deposition of John Lohr, October 11, 2010, TR at 135:16 – 22 (“Q: Now, as you increased clinker production, you’d be consuming more raw materials? A: Yes. Q: Okay. And is that one reason why the quarry permit was seeking an increase in the amount of raw material quarried? A: Yes.”).

²⁵ *See*, Memorandum from John Lohr to David Repasz and Bruce Ballinger, Re: Lyons Kiln Production and Oxygen Use, January 4, 1998, CEMEX_NEIC_00003243 at 3245 (“The kiln has been our bottleneck for most of the plant’s history. We have had raw feed production and finish mill capacity available. The improvements in kiln operation in 1997 reversed that situation. With the combustion chamber improvements and the injection of liquid oxygen, the kiln began to outrun the upstream and downstream systems.... In order to grind all the clinker produced we will need the high efficiency separator.... There are also minor improvements in the raw feed drying, raw feed transfer and clinker cooler that are necessary and would be included in this project.”).

²⁶ NSR Manual, p. A.48.

²⁷ From McVehil-Monnett files: NSR Manual excerpts, p. A.33 – A.34, US00035301, and p. A.43 – A.56, US00033821; Handwritten notes of Keith Baugues, after March 6, 1998, meeting with CDPHE, US00033814 at 33820 (“Follow NSR handout”); Letter from Keith Baugues to Gary Parish, Re: Permitting Strategy for Revisions at the Lyons Southdown Facility: MMA Project Number 786-98, March 25, 1998, CMXLYO00155748 at 155752 (citing “New Source Review Manual” as authority for definition of “modification”); Lyons, CMXLYO00120244 (“[I] Any change in the process or method of operation

a. Baseline Emissions for Collective Changes (Case A, Step 2)

The PSD regulations state that “[i]n general, actual emissions as of a particular date shall equal the average rate, in tons per year, at which the unit actually emitted the pollutant during a two-year period which precedes the particular date and which is representative of normal source operation.”²⁸ This two-year period before the “particular date” has come to be known as the “baseline period” before the change occurred.

i. Actual Emissions

Actual NO_x emissions are estimated from the unit’s actual operating hours, production rates, and types of materials processed, stored, or combusted during the baseline period.²⁹ The preferable way to determine “actual emissions” during the baseline period is to use continuous emission monitoring system (CEMS) data, as the CEMS continuously records actual emissions at the stack when the CEMS is operating. CEMEX has a permit condition requiring the use of a CEMS to measure emissions from its kiln stack.³⁰ This CEMS was in use during the baseline period. I used the monthly CEMS NO_x emission data for the kiln stack (where available) as summarized by CEMEX for the years 1995 – 2000.³¹ When CEMS data were unavailable, I used emissions data CEMEX otherwise reported to the State (1994).³² In either instance, I was

which results in an increase greater than the significant level for any criteria pollutant must go through the prevention of significant deterioration (PSD) review process. [(12) Colorado Department of Health has taken a very firm position as to how this determination is to be made. The future potential-to-emit, based on permit limits and annual hours of operation, must be compared with the average of the actual, not allowable, emissions for the past two years.”).

Deposition of Keith Baugues, July 30, 2010, TR at 80:7 – 25 ([In reference to the handwritten note, “follow NSR Manual”]: “There is a document from 1980 (sic) that is a New Source Review workbook that EPA prepared. It’s a draft, but it’s never been finalized, and that is the guidance that explains how to do emission calculations...the State referred him [Jeet Gill] specifically to this guidance...”); Deposition of Bradley Cooley, September 20, 2010, TR at 26:13 – 15 (“But anyway, the EPA/NSR workshop guidance document was, you know, essentially the document we followed....”); Deposition of Roland Hea, November 2, 2010, TR at 92:6 – 9; Deposition of Hans Buenning, December 8, 2010, TR at 134:22 – 138:3.

²⁸ 40 CFR 52.21(b)(21)(ii); CAQCC Common Provisions, I.G (1986); CAQCC Part A, I.B.1.a (1997).

²⁹ 40 CFR 52.21(b)(21)(ii); CAQCC Common Provisions, I.G (1986); CAQCC Part A, I.B.1.a (1997).

³⁰ CDPHE, Title V Operating Permit, Permit No. 12BO444-2, December 17, 1993, Condition 2, CEMEX_NEIC_00000203.

³¹ CEMEX, NO_x Emissions – Kiln – Historical Summary, CMXLYO00628845 – 628848. I used NO_x emissions without moisture correction for 2000, based on: Deposition of Chuck Kendrick, January 11, 2011, TR at 34:10 – 23; CEMEX Lyons Historic NO_x Emission Trends, CMXLYO00131241 at 131242; Memorandum from Matthew Parks to Paul Carr Re: CEMEX CEMS, September 10, 2007, CEMEX_NEIC_00008623 at 8624 – 8626. My use of CEMEX’s “CEMEX, NO_x Emissions – Kiln – Historical Summary” is done for convenience and should not be construed as an endorsement of those estimates.

³² See, CDPHE Air Pollution Emissions Notice (APEN), December 6, 1995, US00006692 at 6694 (for 1994).

not able to independently confirm CEMEX's CEMS NO_x emissions data because the underlying hourly CEMS data were not produced for most years.

ii. Baseline Period

Actual emissions are determined for the baseline period. This period is the 24 consecutive months preceding the start of the permitting process,³³ standing in the shoes of the permit engineer. This is due to practical considerations; the source needs to look at historical data at the time it prepares its application. Here, CEMEX did not prepare a permit application, and the changes identified in Table 1 were not permitted for NO_x. However, it is feasible to determine the baseline retrospectively as the 24 months prior to construction. To bracket the likely range of the NO_x emissions increase, I evaluated two baseline scenarios for each clinker production goal: (1) the 24 months prior to the start of construction (Scenarios "1" and "2", at 546,250 ton/yr) and (2) the 24 months prior to the year in which permitting should have occurred (Scenarios "3" and "4", at 577,920 ton/yr).

Baseline for Scenarios 1 and 3. The produced documents indicate that the first change, the alteration of the internal configuration of the calciner combustion chamber, occurred during the March 1997 annual maintenance shutdown (See Table 1).³⁴ Thus, the baseline period for the changes taken collectively is the 24-month period from March 1995 through February 1997. The NO_x emissions average for this 24-month period is **1,747 ton/yr.**³⁵ The clinker production average for this 24-month period is **430,741 ton/yr.**³⁶ See Exhibit 1, Tab: Baseline 1&3, Line 48.

³³ 45 Fed. Reg. 52676, 52699, 52705, 52718. *See also*, 61 Fed. Reg. 38259, 38254.

³⁴ Lyons Plant – Monthly Report of Operations, March 1997, CMXLYO00049696 at 49753 ("The kiln outage jobs included several improvements. The combustion chamber modification in the precalciner should improve combustion efficiency by about 2%....").

³⁵ NO_x emissions for the 24-month baseline period, from March 1995 – February 1996, are taken from CEMEX, NO_x Emissions – Kiln – Historical Summary, CMXLYO00628845. I estimated the missing monthly NO_x emission values for March and April 1995 by using the average 1995 NO_x emission factor (lbs of NO_x per ton of clinker) in lb/ton: (1,514.7)(2000)/425,274 = 7.12 lb/ton and the clinker production for each month: (March = 39,669 ton/mo; April = 38,131 ton/mo); *see*, CEMEX, Estimated Clinker Production – Historical Summary, CMXLYO00628849 – 628852. This yields NO_x emissions of 141.3 ton/mo for March 1995 and 135.8 ton/mo for April 1995. I summed the clinker production for each month from March 1995 through February 1997 and divided by 2 to get the annual average baseline clinker production. My use of CEMEX's "NO_x Emissions – Kiln – Historical Summary" is done for convenience and should not be construed as an endorsement of those estimates.

³⁶ Clinker production for the 24-month baseline period, from March 1995 – February 1996, is taken from CEMEX, Estimated Clinker Production – Historical Summary, CMXLYO00628849. The missing monthly values in this summary are taken from Untitled Spreadsheets, CMXLYO00179829 at 179838: October 1995 (32,793 ton/mo); November 1995 (39,040 ton/mo); *see also*, EPA, LYO-19.xls, US0011058 at 111065. My use of CEMEX's "Estimated Clinker Production – Historical Summary" is done for convenience and should not be construed as an endorsement of those estimates.

Baseline for Scenario 2 and 4. The change to the calciner combustion chamber was planned between 1995³⁷ and 1996.³⁸ The design drawings for construction of the change to the calciner combustion chamber were issued October 25, 1996.³⁹ Thus, permitting for this change would have occurred prior to October 1996 to assure a PSD permit was in hand before the March 1997 outage. Because it can take up to a year from application completion to secure a PSD permit, the permitting process would have commenced in early 1996.

Standing in the shoes of a permit engineer at the time of permitting, the baseline period most likely would have been the 24 months prior to early 1996. Because CEMEX has not produced monthly data prior to 1995, and as the start date for permitting cannot be precisely determined retrospectively, I used the two years prior to early 1996, or 1994 – 1995, to establish baseline conditions for Scenarios 2 and 4. Standing in the shoes of a permit engineer at the time the project was contemplated, the average NOx emissions for 1994 – 1995 would have been **1,605 ton/yr**,⁴⁰ and the average clinker production for this two-year period would have been **421,890 ton/yr**.⁴¹ See Exhibit 1, Tab: Baseline 2&4, Line 48.

The resulting baseline clinker production and NOx emissions for these four scenarios, calculated in the immediately preceding footnotes, are summarized in Table 2. This table shows that the baseline NOx emissions for the changes identified in Table 1, taken collectively, ranges from **1,605 ton/yr** to **1,747 ton/yr**.

³⁷ Deposition of John Lohr, October 11, 2010, TR at 52:8 – 12.

³⁸ Case Study of Physical Modeling at Southdown's Lyons Plant, CMXLYO00402635 – 402640.

³⁹ Existing & Modified Calciner Vessel, Drawing Issued for Construction, October 25, 1996, CEMEX_NEIC_00007244.

⁴⁰ For 1994, the annual NOx emissions were 1,695.8 ton/yr, based on CDPHE, APEN, December 6, 1995, US00006692 at 6694, and Lyons Kiln History and Forecast, US00000046. For 1995, the annual NOx emissions were 1,514.7 ton/yr, based on CEMEX, NOx Emissions – Kiln – Historical Summary, CMXLYO00628845. The average of these two values is **1,605 ton/yr**. My use of CEMEX's "NOx Emissions – Kiln – Historical Summary" is done for convenience and should not be construed as an endorsement of those estimates.

⁴¹ For 1994, the annual clinker production was 418,505 ton/yr, based on CDPHE, APEN, December 6, 1995, US00006692 at 6694, and Lyons Kiln History and Forecast, US00000046. For 1995, the annual clinker production was 425,274 ton/yr, based on CEMEX, Estimated Clinker Production – Historical Summary, CMXLYO00628849. The average of these two values is **421,890 ton/yr**. My use of CEMEX's "Estimated Clinker Production – Historical Summary" is done for convenience and should not be construed as an endorsement of those estimates.

Table 2:
Baseline Clinker Production and NOx Emissions
Case A, Step 2

	Clinker Production (ton/yr)	NOx Emissions (ton/yr)	NOx Emission Factor ⁴² (lb/ton)
Scenarios 1 and 3	430,741	1,747	8.11
Scenarios 2 and 4	421,890	1,605	7.61

b. Potential to Emit for Collective Changes (Case A, Step 2)

The PSD regulations also state that “[f]or any emissions unit which has not begun normal operations on the particular date, actual emissions shall equal the potential to emit of the unit on that date.”⁴³ EPA requires a source to predict its post-change “potential to emit.” The potential to emit is the capability of a source at maximum design capacity to emit a pollutant, except as constrained by federally enforceable conditions.⁴⁴ Potential to emit is the appropriate method to estimate future emissions from the subject collective changes at the time they should have been permitted.⁴⁵

A commonly used method for calculating potential to emit NOx from a cement kiln is to multiply a pre-change emission factor expressed in pounds of NOx per ton of production (*e.g.*, clinker, cement, raw feed) by the maximum production allowed by the change.⁴⁶ CEMEX used emission factors to determine compliance before the State mandated the use of CEMS.⁴⁷

As explained below, two pieces of information are required to complete this calculation: (1) the maximum potential clinker production and (2) a NOx emission factor in lb/ton. Thus, an increase in production alone, by simple arithmetic,⁴⁸ increases NOx

⁴² Calculated as: (NOx emissions in ton/yr) (2000 lb/ton)/(clinker production in ton/yr) = NOx emission factor in lb/ton.

⁴³ 40 CFR 52.21(b)(21)(iv); CAQCC Common Provisions, I.G (1986); CAQCC Part A, I.B.1.c (1997).

⁴⁴ 40 CFR 52.21(b)(4) (1995) (presently recodified as 40 CFR 52.21(b)(23)(i)); CAQCC Common Provisions, I.G (1986); CAQCC Part A, I.B.44 (1997). *See also*, NSR Manual, p. A.1 and Sec. II.B, p. A.4 – A.22.

⁴⁵ *See*, NSR Manual, p. A.39 and A.41; Deposition of Keith Baugues, July 30, 2010, TR at 249:11 – 15.

⁴⁶ *See*, Emission factors in U.S. EPA, AP-42, Section 11.6, Portland Cement Manufacturing, Table 11.6 – 8. *See also*, Chapter 4, Uncontrolled NOx Emissions, CMXLYO00490929.

⁴⁷ Boulder County Health Department (BCHD) Air Inspection Report, September 5, 1995, US00005590 at 5596.

⁴⁸ Deposition of Andrew Jackura, September 16, 2010, TR at 116:7 – 10; *United States v. CEMEX California Cement, LLC.*, U.S. District Court for the Central District of California, Western Division, Civil Action cv-07-00223-GW, Deposition of Amarjit Singh Gill, March 18, 2008, TR at 362:21 – 363:13 (“Q: Now, does the day-in/day-out use of a constant NOx emissions factor—does it assume that there is a

emissions. The following sections first discuss calculating the increase in clinker production resulting from the collective changes and then address calculation of a NOx emission factor.

i. Maximum Potential Clinker Production for the Collective Changes
(Case A, Step 2)

A range of clinker/cement production targets for the collective changes is found in the record.⁴⁹ In its final Approval for Expenditure (AFE), CEMEX projected 575,000 ton/yr of cement⁵⁰ as a result of the changes identified in Table 1; in its 1998 and 1999 capital budgets implementing the AFE, CEMEX projected up to 1720 ton/day of clinker because of these changes.⁵¹ This amounts to, respectively, 546,250 ton/yr⁵² to 577,920 ton/yr⁵³ of clinker production. The baseline clinker production ranges from 430,741

relationship, then, between production and emissions? A: No. Q: So, the emissions factor appears to stay constant in your table. A: Yes. Q: So, how is it that there is not—I mean, all you’re doing is saying emissions factor times production equals emissions. A: That is correct. Q: So, in that calculation, as production goes up, emissions go up. Is that correct? A: That is correct.”).

⁴⁹ Memorandum from Dave Repasz to Bruce Tompkins, Re: Lyons Visit, **October 26, 1997**, CEMEX_NEIC_00003281 at 3282 (600,000 to 840,000 ton/yr of cement in two phases); Lyons Plant Expansion – Phase I, Project Scope, Revised **December 12, 1997**, US00000152 (600,000 ton/yr of cement); Memorandum from John Lohr to Dave Repasz and Bruce Ballinger, Re: Lyons Kiln Production and Oxygen Use, **January 4, 1998**, US00000020 (“This permit is not expected to be a problem as we push clinker production up to 550,000 tons/year.”); Handwritten notes of Ram Seetharam, **January 6, 1998**, Seetharam Ex. 61, US00006344 (“clinker/cement target 600,000 in 1998[;] 700,000 in 1999.”); Memorandum from Gregg St. Clair to John Lohr, **January 15, 1998**, Re: Plant Expansion Capacity Limitations, US00044353 (list by year of recommended capital improvements “for expanding the plant’s capacity”. The 1998 cement production goal is 525,000 ton/yr; the 1999 cement production goal is 575,000 ton/yr, the 2000 cement production goal is 850,000 ton/yr); Authorization for Expenditure, Lyons Capacity Increase, Project Scope, Revised **February 20, 1998**, CEMEX_NEIC_00003335 at 3337 (“The proposed project will increase capacity by 125,000 tons to 575,000 tons per annum....”); Memorandum from Karen Krenek to Norris Widener, Re: Approved AFE, **November 18, 1998**, CEMEX_NEIC_00003449 at 3452 (“The proposed project will increase capacity by 125,000 tons to 575,000 tons per annum.... A portion of this capacity increase was achieved in 1997 through minor modifications to the preheater aerodynamics combined with the temporary use of high cost purchased oxygen.”) and at 3457 (“The plant expansion addresses the system components that need to be modified or replaced so that plant capacity will increase to 575,000 tons per year of cement.”).

⁵⁰ Memorandum from Karen Krenek to Norris Widener, Re: Approved AFE, November 18, 1998, CEMEX_NEIC_00003449 at 3452 and 3457 (575,000 ton/yr of cement).

⁵¹ 1998 Capital Budget, Item 6 – Lyons Capacity Increase, May 9, 2007, US00044356 (“Anticipated Benefits: Support kiln production rates up to 1720 tpd and equivalent seasonalized cement production.”); 1999 Capital Budget, Item 2 – Lyons Capacity Increase, August 28, 1998, CMXLYO00155783 at 155787 (“Anticipated Benefits: Support kiln production rates up to 1720 tpd and equivalent seasonalized cement production.”).

⁵² The ratio of clinker to cement for the Lyons Plant is 0.95 ton of clinker per ton of cement. *See, e.g.*, Southdown, Inc., Lyons Cement Plant, Clinker/Cement Production, US00000583 – 584). Thus, the cement production goal of 575,000 ton/yr requires clinker production of: (575,000)(0.95) = **546,250 ton/yr**.

⁵³ The 1997 – 2000 kiln permit (CDPHE – APCD, Construction Permit, Permit No. 12BO444-2, December 17, 1993, Condition 1, CEMEX_NEIC_00000203) and the current operating permit (CDPHE, Title V Operating Permit, Last Revised October 8, 2002, US00121584 at 121621) limit the kiln operating hours to

ton/yr for Scenarios 1 and 3 (March 1995 to February 1997) to 421,890 ton/yr for Scenarios 2 and 4 (1994 – 1995) (Table 2). Thus, the increase in clinker production allowed by these changes ranges from **115,509 ton/yr** to **156,032 ton/yr**.⁵⁴

The changes to the pyroprocessing system (combustion chamber, ID fan motor, oxygen injection, and annulus boost fan) increased the throughput of the system.⁵⁵ Other projects debottlenecked the upstream and downstream processes (*e.g.*, raw mill, finish mill, and quarry) to take advantage of the increased capacity of the pyroprocessing system.⁵⁶

A bottleneck is a limit on the production at a given point due to restrictions at upstream or downstream units that prevents a system from reaching its full capacity.⁵⁷ If

8,064 hr/yr. Thus, the maximum potential clinker production is: (1,720 ton/day)(365 day/yr)(8,064hr/yr/8,760 hr/yr) = **577, 920 ton/yr**.

⁵⁴ Increase in clinker production ranges from: (546,250 – 430,741) = **115, 509 ton/yr** to (577,920 – 421,888) = **156,032 ton/yr**.

⁵⁵ Expert Report of Walter L. Greer, p. 26 – 32, 35 – 36.

⁵⁶ Expert Report of Walter L. Greer, p. 33 – 34, 37 – 40.

⁵⁷ See also, Deposition of John Lohr, October 11, 2010, TR at 140:25 – 141:6 (“Bottleneck, to me, is something that will limit or has the potential to limit our production, and I believe that that particular term is used in EPA regulations...”), at 191:6 – 11 (“...It means eliminating the limitations.”);

Deposition of Keith Baugues, July 30, 2010, TR at 13:9 – 25 (“Q: This memo refers to the kiln as a bottleneck. A: Yes. Q: What does that term mean to you? A: I guess in the term of air pollution, the throughput going to this particular unit as hampered by the kiln. In other words, it could have done—the entire overall process could have made more material if the kiln could have operated at a higher capacity or better throughput. Something in the kiln was not operating correctly to give them an increased input. Something they must have changed in ’97 changed that, therefore, as I say, they could produce more tons of cement per hour.”);

Deposition of Ram Seetharam, August 11, 2010, TR at 23:9 – 24:11 (“Q: Is this what’s known as debottlenecking? A: See, if a facility has several—several processes, say, and then one is dependent on the previous, that one is dependent on the previous one, and some of these sub processes may have different design rates or production capacities and some may have much lower than those—the other processes, so when—when the facility decides that this particular one has become a constraint for their—for their entire facility, if they decide to do a modification of that particular process, make some changes in the equipment, make some maybe – maybe an augmentation of the particular one to sort of increase – increase a particular one, this process is known as debottlenecking, in a sense, that would be subject to modification reviews...the facility itself, its potential to emit have increase because of this particular debottlenecking.”);

Deposition of David Repasz, September 15, 2010, TR at 47:14 – 17 (“Q: First of all, what’s a bottleneck? A: It’s a limit on a process. Physical limit, I’d say.”);

Deposition of Daniel Heintz, September 21, 2010, TR at 35:17 – 20 (“Well, what it means to me is undertaking some project or work to alleviate some sort of backup in a system so that you can process more material or make more product or something like that.”), and at 37:17 – 23 (“I mean, if—if a project did debottleneck a process or an operation, it might require a review as to whether or not a permit is required.”);

Deposition of Bradley Cooley, September 9, 2010, TR at 58:10 – 18 (“Q: Do you have an understanding of what the term ‘bottleneck’ means? A: Yes. Q: Can you tell me what that is? A: Yes, it’s a limitation in production at a point in the plant. Q: And if that limitation in production is removed, is there a term for that? A: Debottlenecking.”).

the limitation is removed, the process is said to be debottlenecked.⁵⁸ The “capacity” of a debottlenecked unit, for purposes of PSD permitting, is determined by the “but for” test. If an increase could not occur “but for” a change required elsewhere in the facility, removing that limitation is debottlenecking.⁵⁹ As noted previously, the increase in emissions that results from removing bottlenecks is included in emission calculations used to determine PSD applicability.

As explained in deposition testimony and Mr. Greer’s Expert Report, producing more clinker requires burning more fuel, which generates more NOx.⁶⁰ In other words, as clinker production goes up, NOx emissions go up.⁶¹ As Mr. Greer put it: “There is no reason to believe that increased NOx would not have been emitted to atmosphere in about the same amount relative to clinker production as was emitted in the year prior to the modification.”⁶² Therefore, for purposes of calculating the potential to emit NOx in Step 2, the sum of the increases in clinker production allowed by debottlenecking the kiln, including the upstream and downstream units (raw mill, finish mill, and quarry), must be included.

⁵⁸ See, e.g., Deposition of Bradley Cooley, September 9, 2010, TR at 58:10 – 18; NSR Manual, p. A.37, A.46.

⁵⁹ See, e.g., NSR Manual, p. A.37, A.46; see also, e.g., Letter from R. Douglas Neeley, EPA Region 4, to Rhonda Banks Thompson, South Carolina Department of Health and Environmental Control, Re: Request for Guidance on PSD Applicability Determinations for Boiler Emissions, March 14, 1997, available at: <http://www.epa.gov/region7/air/nsr/nsrmemos/hoechst.pdf> (discussing the “but for” test); Patrick W. Foley, EPA Region 3, Debottlenecking Slides, 1999, available at: <http://daq.state.nc.us/permits/psd/docs/db1.pdf>; Letter from John Daniel, EPA Region 3, Re: Intermet Corp., October 23, 13, 1998, available at: <http://www.epa.gov/ttn/nsr/techinfo.html>; Letter from Sam Portanova, EPA Region 5, to Steve Dunn, Wisconsin Dept of Natural Resources, February 24, 2005, Re: Request for a PSD Applicability Determination for *Murphy Oil*, available at: <http://yosemite.epa.gov/r5/r5ard.nsf/2134f82000aa062c86257577004df4d7/f7477350acb2d7ad862574c8006fd129!OpenDocument>.

⁶⁰ Expert Report of Walter L. Greer, p. 22, n. 88 and 23. Deposition of Andrew Jackura, September 16, 2010, TR at 116:7 – 10 (“Q: If the emission rate as gauged in pounds per ton of clinker remained constant, if one produces more clinker, does one, by necessity, produce more NOx? A: Yes. Per – over a given time period, yes.”); Deposition of Ram Seetharam, August 11, 2010, TR at 58:5 – 59:11, 75:7 – 77:25 (“...A: Yes, it suggests that, yes, there is an increase. Q: And that would be an increase in throughput? A: Yes, that’s right. Q: Would this have had the potential to increase NOx emissions? A: Yes, whatever, but other pollutants too...”), 107:15-20, 110:7-9; Deposition of David Repasz, September 15, 2010, TR at 32:24 – 33:7. ERG Inc., Chapter 3.0, “NOx Formation in Cement Kilns”, *Assessment of NOx Emissions Reduction Strategies for Cement Kilns – Ellis County*, Final Report, July 14, 2006; see, e.g., p. 3 – 11 (“Since NOx formation is directly related to fuel combustion, any reduction in the amount of fuel burned per unit amount of clinker produced should reduce NOx emissions per unit of clinker.”).

⁶¹ *United States v. CEMEX California Cement, LLC.*, U.S. District Court for the Central District of California, Western Division, Civil Action cv-07-00223-GW, Deposition of Amarjit Singh Gill, March 18, 2008, TR at 362:21 – 363:13 (“Q: Now, does the day-in/day-out use of a constant NOx emissions factor— does it assume that there is a relationship, then, between production and emissions? A: No. Q: So, the emissions factor appears to stay constant in your table. A: Yes. Q: So, how is it that there is not—I mean, all you’re doing is saying emissions factor times production equals emissions. A: That is correct. Q: So, in that calculation, as production goes up, emissions go up. Is that correct? A: That is correct.”).

⁶² Expert Report, of Walter L. Greer, p. 27.

ii. NOx Emission Factor for the Collective Changes (Case A, Step 2)

Next, maximum potential clinker production is multiplied by a NOx emission factor in pounds of NOx per ton of clinker to determine the potential to emit NOx. This section discusses the selection of the NOx emission factor used to calculate the potential to emit for the changes taken as a whole. The following section presents the potential to emit calculation using this factor.

A recognized method for calculating NOx emissions from a cement kiln is to multiply an emission factor expressed in pounds of NOx per unit of production by the production goal.⁶³ I estimated the potential to emit NOx using pounds of NOx per ton of clinker (lb/ton), based on the available data. A permit engineer making this calculation during the baseline period would have followed this approach. For example, a permit engineer retained by CEMEX estimated the potential to emit from the anticipated increase in raw feed and the pounds of NOx per ton of raw feed, based on 1997 CEMS data.⁶⁴

I used the baseline NOx emission factors as summarized in Table 2 for the two baseline scenarios. For Scenario 1 and 3 (March 1995 to February 1997) this works out to an emission factor of 8.1 lb/ton. For Scenarios 2 and 4 (1994 – 1995), the baseline standing in the shoes of a permit engineer, this works out to an emission factor of 7.6 lb/ton. (See Table 2.) A permit engineer in 1994 – 1995 would have followed this method of calculating potential to emit, namely, multiplying a constant baseline NOx emission factor by the anticipated increase in production.

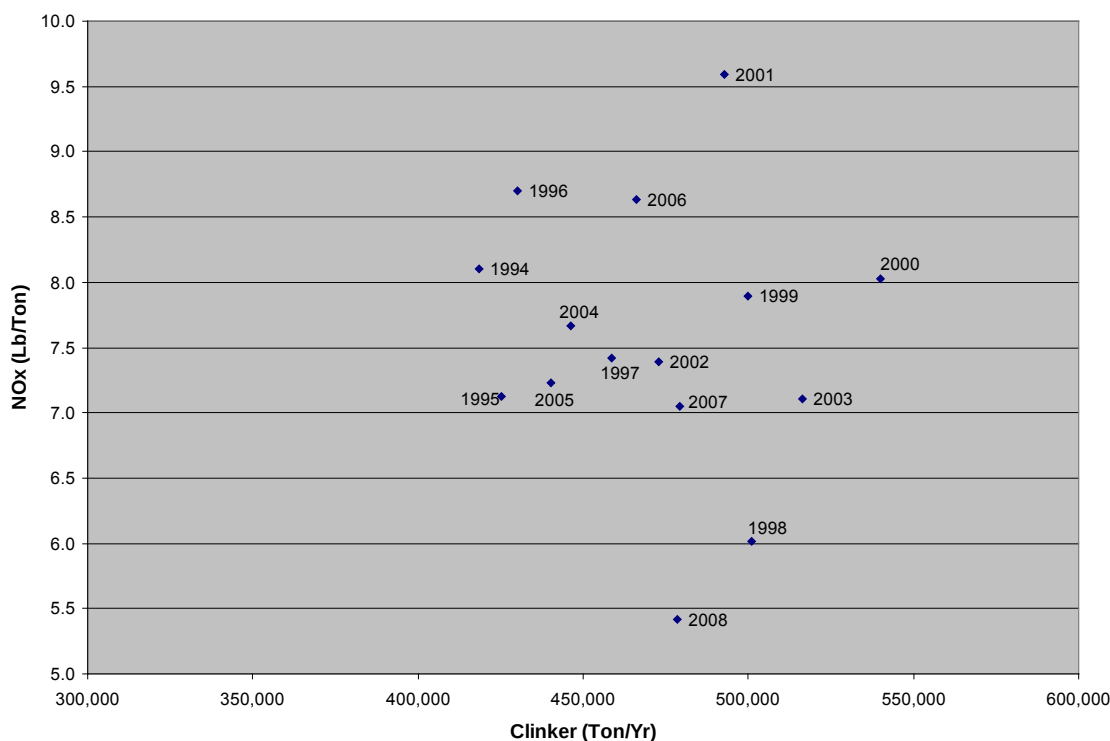
I also evaluated the relationship between the NOx emission factor and clinker production for the Lyons kiln retrospectively by plotting the actual annual NOx emissions in pounds of NOx per ton of clinker against actual annual tons of clinker. Figure 3 shows that NOx emissions from the Lyons kiln are inherently variable, consistent with results reported for other kilns.⁶⁵ These data indicate that, within the limits of variability and accuracy of measurement, the NOx emission factor is constant over the range of clinker production.

⁶³ U.S. EPA, AP-42, Section 11.6 Portland Cement Manufacturing, Table 11.6-8; EPA, Alternative Control Techniques Document – NOx Emissions from Cement Manufacturing, Report EPA-453/R-94-004, March 1994.

⁶⁴ Fax from Keith Baugues to John Lohr, March 11, 1998, attaching handwritten section on “Kiln and Dry Emissions,” US0000065 at 80 – 82, US00000062 (conclusions), and US00000090 (Table 1 handwritten); US00034559 at 34562 (fax cover sheet) and 34618 – 34620 (memo); and US00034473 at 34474 (Table 1 typewritten). Letter from Keith Baugues to Bradley Cooley, Re: Revised Emission Spreadsheets: MMA Project 786-98, March 26, 1998, US00035294.

⁶⁵ Walter L. Greer and Curtis D. Lesslie, “Variability of NOx Emissions from Precalciner Cement Kiln Systems”, *PCA R&D Serial No. 2782*, 2001; L.J. Walters, Jr., et al., “Time-Variability of NOx Emissions from Portland Cement Kilns”, *Environmental Science & Technology*, v. 33, 1999, p. 700 – 704.

Figure 3:
Annual NOx Emission Factor (lb/ton) v. Clinker Production (ton/yr)⁶⁶



Curve fitting plus an analysis of variance⁶⁷ indicates no statistically significant relationship between the NOx emission factor in lb/ton and clinker production in ton/yr. The average NOx emission factor for the period of record, 1994 – 2008, is 7.6 lb/ton. The average NOx emission factor for the baseline periods I have identified for Case A ranges from 7.6 lb/ton to 8.1 lb/ton (Table 2). Thus, there is no basis to assume that modifications to the Lyons Plant changed the relationship between NOx and clinker production over the period of record.

⁶⁶ The data plotted in this figure is derived from CEMEX documents as follows: (1) annual clinker production for 1995 – 2008 is from CEMEX, Estimated Clinker Production – Historical Summary, CMXLYO00628849 – 628852. Annual clinker production for 1994 is from CDPHE, APEN, December 6, 1995, US00006692 at 6694. (2) Annual NOx emissions for 1995 – 2008 are from CEMEX, NOx Emissions – Kiln – Historical Summary, CMXLYO00628845 – 628848. Annual NOx emissions for 1994 are from CDPHE, APEN, December 6, 1995, US00006692 at 6694. I used NOx emissions without moisture correction for 2000 – 2001, based on: Deposition of Chuck Kendrick, TR at 34:12-23; Memorandum from Matthew Parks to Paul Carr, Re: CEMEX CEMS, September 10, 2007, CEMEX_NEIC_00008623 at 8624 – 8626. NOx emissions expressed in lb/ton are calculated as: (NOx emissions ton/yr)(2000 lb/ton)/(clinker production ton/yr). My use of CEMEX’s “Estimated Clinker Production – Historical Summary” and “NOx Emissions – Kiln – Historical Summary” is done for convenience and should not be construed as an endorsement of those estimates.

⁶⁷ For analysis of variance calculations, see Exhibit 2 (Appendix D) attached to my report.

iii. Potential to Emit NOx for Collective Changes (Case A, Step 2)

Using the above future potential clinker production range and the NOx baseline emission factor, the potential to emit NOx ranges from **2,076 ton/yr** to **2,341 ton/yr**, as demonstrated in Table 3.

Table 3:
Potential to Emit NOx Due to Collective Changes Made Between 1997 – 2000
Case A, Step 2

Figure 1	Future Potential Clinker Production (ton/yr)	NOx Emission Factor (lb/ton)	Potential to Emit NOx ⁶⁸ (ton/yr)
Scenario 1	546,250	8.1	2,212
Scenario 2	546,250	7.6	2,076
Scenario 3	577,920	8.1	2,341
Scenario 4	577,920	7.6	2,196

2. NOx Emission Increase for Collective Changes (Case A, Step 2)

The increase in NOx emissions resulting from the collection of changes in Step 2 is the difference between the potential to emit NOx in Table 3 and the baseline NOx emissions in Table 2. This calculation is shown in Table 4.

Table 4:
Increase in NOx Due to Collective Changes Made Between 1997 – 2000
Case A, Step 2

Figure 1	Potential to Emit NOx (ton/yr)	Baseline NOx Emissions (ton/yr)	Increase in NOx ⁶⁹ Emissions (ton/yr)
Scenario 1	2,212	1,747	465
Scenario 2	2,076	1,605	471
Scenario 3	2,341	1,747	594
Scenario 4	2,196	1,605	591

The NOx emission increase related to the changes made at the Lyons Plant between 1997 – 2000 ranges from **465 ton/yr** to **594 ton/yr**, depending on the baseline

⁶⁸ Potential to emit NOx = future clinker production x NOx emission factor x 2000 lb/ton.

⁶⁹ Increase in NOx emissions = potential to emit NOx – baseline NOx emissions.

period and CEMEX's cement production goals. These values exceed the significance threshold of 40 tons/yr. Thus, Step 3 netting is required to determine if PSD applies.⁷⁰

My estimate of the NOx emission increase in Table 4 is consistent with estimates made by McVehil-Monnett in 1998. Their estimates ranged from 452 ton/yr⁷¹ (1995 – 1996 baseline) to 324 ton/yr⁷² (1996 – 1997 baseline) for 575,000 ton/yr of clinker. I found no evidence in the documents produced to date or in deposition testimony that CEMEX rejected the McVehil-Monnett calculations.⁷³ The differences between these various estimates are immaterial and do not affect my opinion that the collective changes would have resulted in an increase in NOx emissions of a sufficient magnitude to trigger PSD review.

3. Netting Analysis for Collective Changes (Case A, Step 3)

A major modification is any physical change in or change in the method of operation of a major stationary source that would result in a significant emissions increase by itself and a significant net emissions increase.⁷⁴ Section IV.B.2 demonstrates that the changes in Table 1, viewed collectively, would result in a significant emission increase, satisfying the first part of this test.

⁷⁰My calculations do not include increased NOx emissions from the dryer from increased raw feed as these NOx increases are small, relative to the NOx increase from the pyroprocessing system. *See, e.g.*, Letter from Keith Baugues to John Lohr, Re: Kiln and Dryer Emission Estimates: MMA Project Number 786-98, March 12, 1998, CMXLYO00038872 at 38874.

⁷¹ Emissions Summary (TPY), 1995 – 96 Baseline, US00028915 at 28920 (786-98/Keith/Summary.xls) and US00000047 at 51 (**451.93 ton/yr**).

⁷² Emissions Summary (TPY), 1996 – 97 Baseline, US00028915 at 28921 (786-98/Keith/Summary.xls) and US00000047 at 52 (**323.63 ton/yr**); Fax from Keith Baugues to John Lohr, March 11, 1998, attaching handwritten section on “Kiln and Dry Emissions,” US00000065 at 80 – 82, and US00000090 (Table 1 handwritten, **365 ton/yr**); Letter from Keith Baugues to John Lohr, Re: Kiln and Dryer Emission Estimates: MMA Project Number 786-98, March 12, 1998, CMXLYO00038872 at 38874 (**365 ton/yr**).

⁷³ Deposition of Keith Baugues, July 30, 2010, TR at 250:9 – 16 (“Q: Okay, With respect to the netting analysis, particularly as to the kiln and dryer, did anyone at the plant ever tell you that that analysis was erroneous? A: No. Did they accept that analysis as being a valid analysis? A: Yes.”).

John Lohr testified that Jeet Gill criticized the Baugues analysis as being too “simplistic.” Mr. Lohr incorrectly testified the analysis was based on AP-42 emission factors when, for the kiln, it was based on actual Lyons CEMS emissions data, the most accurate way to estimate emissions. Deposition of John Lohr, October 11, 2010, TR at 184:16 – 185:16, 186: 13 – 15, and 187:3 – 15.

Deposition of William Monnett, November 30, 2010, TR at 33:15 – 24, 51:24 – 52:9 ([Jeet Gill never informed McVehil-Monnett that their netting analysis was wrong]: “...I recall him saying at a meeting, I don’t care what your opinion is, I have mine.”).

Deposition of Bradley Cooley, TR at 37:16 – 19, 97:18 – 20 (“Q: Did you ever hear anyone at CEMEX criticize any netting analysis done by McVehil-Monnett? A: No, not specifically.”), 98:8 – 10.

⁷⁴ *See*, NSR Manual, P. A.33 – A.34.

Step 3 requires a determination of the net change in emissions. If both the emission increase in Step 2 and net change in emissions in Step 3 equal or exceed 40 ton/yr, PSD review is required.⁷⁵

The net emission change equals the emission increase from Step 2, minus source-wide creditable and contemporaneous emission decreases, plus source-wide creditable and contemporaneous emission increases.⁷⁶ Collectively, these two steps are referred to as “netting.” The contemporaneous period for the collective changes made between 1997 and 2000 is 5 years prior to the start of construction of the first change, modification of the calciner combustion chamber in March 1997, through startup of the last change, the addition of the annulus boost fan in August 2000.⁷⁷ I am not aware of any other contemporaneous and creditable emission increases or decreases that occurred during this period.⁷⁸ Thus, Step 2 emissions equal Step 3 emissions.

As net emission increases in the range of 465 ton/yr to 594 ton/yr exceed the PSD significance threshold for NO_x, the changes considered collectively are a major modification of a major stationary source. PSD review would have been triggered and a PSD permit should have been obtained prior to the start of construction in March 1997.

C. Case B: Individual Changes

I separately assessed the impact on emissions for each change identified in Table 1, following the same general procedures outlined above in Section IV.B (Fig. 2). As demonstrated below, when each change is evaluated separately for PSD applicability, the same conclusion is reached as for Case A, *i.e.*, that a PSD permit was required prior to the start of construction for each change.

1. NO_x Emission Increase for Individual Changes (Case B, Step 2)

In Case A, it was possible to estimate the point at which permitting for the collective changes would most likely have occurred, based on the produced documents. However, in Case B, due to gaps in the record it is not possible to make this determination for each change evaluated individually. The start construction date for Case A can be ascertained and is summarized in Table 1. However, for Case B, the date that permitting would have started for each change taken individually cannot be ascertained. Thus, for Case B, as it can take up to a year from application completion to secure a PSD permit, I assumed permitting would have commenced at least 1 year prior

⁷⁵ NSR Manual, p. A.35, A.46.

⁷⁶ NSR Manual, Sec. III.B., p. A.34 – A.55.

⁷⁷ Email from John Lohr to Steve Bryan, Re: Lyons Kiln Status, August 17, 2000, CMXLYO00157315.

⁷⁸ *See also*, Expert Report of Walter L. Greer, p. 33, n. 164.

to the start of construction for all changes except Change #4.⁷⁹ The two scenarios in Case B bracket the likely baseline conditions.

For each individual change in Table 1, I calculated the increase in clinker production, NOx emissions, and the NOx emission factor for the baseline period (the 24 months prior to the start of construction). The data I used to determine the baseline clinker and NOx are as reported by CEMEX.⁸⁰ The calculations are in Exhibit 1. The results of these calculations are summarized in Table 5 for Scenario 1 (as of the start of construction) and in Table 6 for Scenario 2 (standing in the shoes of a permit engineer). These baselines are used below in Tables 7 and 8 to calculate the increase in NOx emissions from each change.

⁷⁹ The larger ID fan motor was installed in August 1997. Thus, the baseline standing in the shoes of a permit engineer is the same as the baseline as of the start of construction. However, the decision to run the motor at a higher rotational speed than the former motor could have and should have been anticipated and permitted. See, Letter from John Lohr to Emergency Breakdown Team, August 11, 1997, CMXLYO00453897.

⁸⁰ CEMEX, Estimated Clinker Production – Historical Summary, CMXLYO00628849 – 628852 and CEMEX, NOx Emissions – Kiln – Historical Summary, CMXLYO00628845 – 628848. The NOx emissions “without moisture correction” were used as discussed in note 30. The monthly clinker production for the period January 1994 through December 1994 is from Clinker Production, Monthly, US00000583. The monthly NOx emissions for the period January 1994 to April 1995 were determined by multiplying the monthly clinker production by a NOx emission factor determined from annual NOx emissions and annual clinker production, viz, for 1994: (1695.8 ton/yr)(2000 lb/ton)/418,505 ton clinker/yr) = **8.10 lb NOx/ton clinker**. For 1995: (1514.7 ton/yr)(2000 lb/ton)/(425,274 ton/yr clinker) = **7.12 lb NOx/ton clinker**.

Table 5:
Baseline Clinker Production and NOx Emissions for Changes Considered
Individually (As of the Start of Construction)
Case B, Step 2, Scenario 1

	Project Name	Clinker Production (ton/yr)	NOx Emissions (ton/yr)	NOx Emission Factor ⁸¹ (lb/ton)
1	Combustion Chamber Modifications	430,741	1,747	8.11
2	Raw Mill Transfer Line	417,794	1,694	8.11
3	Purchased Oxygen Injection	417,794	1,694	8.11
4	ID Fan 1500 HP Motor	433,615	1,739	8.02
5	New Raw Feed Compressor	441,259	1,794	8.13
6	New High Efficiency Separator	470,503	1,627	6.91
7	Quarry	470,503	1,627	6.91
8	Tempering Tee Damper	479,851	1,604	6.69
9	VSA Oxygen Plant	490,658	1,655	6.75
10	New Annulus Boost Fan	514,610	1,906	7.41

Table 6:
Baseline Clinker Production and NOx Emissions for Changes Considered
Individually (Standing in the Shoes of a Permit Engineer)
Case B, Step 2, Scenario 2

	Project Name	Clinker Production (ton/yr)	NOx Emissions (ton/yr)	NOx Emission Factor ⁸² (lb/ton)
1	Combustion Chamber Modifications	421,890	1,605	7.61
2	Raw Mill Transfer Line	407,915	1,529	7.50
3	Purchased Oxygen Injection	407,915	1,529	7.50
4	ID Fan 1500 HP Motor	433,615	1,739	8.02
5	New Raw Feed Compressor	427,013	1,682	7.88
6	New High Efficiency Separator	437,680	1,788	8.17
7	Quarry	437,680	1,788	8.17
8	Tempering Tee Damper	444,501	1,787	8.04
9	VSA Oxygen Plant	463,017	1,831	7.91
10	New Annulus Boost Fan	490,779	1,699	6.92

⁸¹ NOx emission factor = [NOx emissions x 2000 lb/ton]/clinker production.

⁸² NOx emission factor = [NOx emissions x 2000 lb/ton]/clinker production.

Next, I calculated the increase in clinker production for each change. There are two types of changes: (1) changes to the pyroprocessing unit that directly increase NOx emissions and (2) changes to the material handling systems that indirectly increase NOx emissions by removing production limitations or bottlenecks. The debottlenecking changes indirectly allow NOx emissions to increase by increasing the throughput of upstream and downstream units. The increases in NOx emissions shown for these changes occur in the pyroprocessing unit. I used Mr. Greer's estimate of clinker production increases for the individual pyroprocessing changes. I estimated the increase in clinker production for the debottlenecking changes using the production goals reported in CEMEX's AFE⁸³ and documents showing increases in quarry permit limits.⁸⁴ These changes are summarized and estimated as shown in Table 7.

⁸³ Memorandum from Karen Krennek to Norris Widener, Re: Approved AFE, November 18, 1998, CEMEX_NEIC_00003449; Lyons Capacity Expansion, Project Scope, Revised September 21, 1998, CEMEX_NEIC_00003442.

⁸⁴ CDPHE – APCD, Construction Permit, Permit No. 12 BO444-2, December 17, 1993, CEMEX_NEIC_00000203; Lyons Capacity Expansion, Project Scope, Revised February 20, 1998, CEMEX_NEIC_00003449 at 3457.

Table 7:
Increase in Clinker Production for Changes Considered Individually
Case B, Step 2

	Project Name	Type Of Change	Increase In Clinker (ton/yr)	Source
1	Combustion Chamber Modifications	Pyro	23,386	Greer, p. 26
2	Raw Mill Transfer Line	Debottle	*	See Change #8
3	Purchased Oxygen Injection	Pyro	45,158	Greer, p. 30
4	ID Fan 1500 HP Motor	Pyro	12,902	Greer, p. 28
5	New Raw Feed Compressor	Debottle	*	See Change #8
6	New High Efficiency Separator	Debottle	61,286	8 ton/hr x 8064 hr/yr x 0.95 ⁸⁵
7	Quarry	Debottle	48,384	(967,680 – 887,040) x 0.6 ⁸⁶
8	Tempering Tee Damper	Debottle	48,384*	10 ton/hr x 8064 hr/yr x 0.6 ⁸⁷
9	VSA Oxygen Plant	Pyro	28,224	Greer, p. 35
10	New Annulus Boost Fan	Pyro	16,128	Greer, p. 36

* Indicates that a change was made to the raw mill. See note 87, below.

Finally, I calculated the potential increase in NOx emissions from each individual change. The potential to emit is the maximum increase in clinker allowed by a change (Table 7), multiplied by the baseline NOx emission factor in lb/ton, as reported in Tables

⁸⁵ The goal of the finish mill changes was to increase its production rate from 74 ton/hr to 82 ton/hr. Lyons Plant Expansion, Project Scope, Revised February 20, 1998, CEMEX_NEIC_00003449 at 3457. Thus, the debottlenecked clinker production is: 8 ton/hr x 8,064 hr/yr x 0.95 ton clinker/ton cement = **61,286 ton/yr**.

⁸⁶ The permits indicate that the changes at the quarry increased the raw feed sent to the kiln from 887,040 ton/yr (CDPHE –APCD, Construction Permit, Permit No. 12BO444-2, December 17, 1993, CEMEX_NEIC_00000203 at 204) to 967,680 ton/yr (CDPHE, Title V Operating Permit, Last Revised October 8, 2002, US00121584 at 121621). The resulting increase in clinker production: (967,680 – 887,040) x 0.6 = **48,384 ton/yr clinker**. The value 0.6 is the yield factor, which is the fraction of the raw feed that is converted into clinker. Memorandum from Gregg St. Clair to Dave Repasz, March 26, 1998, Re: Plant Expansion Project Update, CMXLYO00155748 at 155750 (Repasz Exhibit 132). This table shows that the historic yield factor ranged from 0.56 to 0.62 between 1992 and 1997 and averaged 0.59, which rounds up to 0.6, the value I used. Further, the second set of production figures on this page is captioned “Anticipated Mix and Dowe Flats Performance after Improvements.” This table shows that CEMEX assumed a yield factor of 0.6 in estimating clinker production for the collection of changes that I am evaluating.

⁸⁷ At least three changes were made to the raw mill (as indicated by asterisk in Table 7): an increase in the diameter of the raw mill transfer line (Southdown, Inc., Fixed Assets Input Request, May 1, 1997 CMXLYO00089625 at 89627), a new raw feed compressor (Lyons Plant Expansion, Project Scope, Revised September 21, 1998, CMXLYO00155865 at 867), and a tempering tee damper (Lyon Plant Expansion – Authorized Budget, November 4, 1998, CEMEX_NEIC_00003449 at 3451). Due to gaps in the record, it is not possible to attribute a specific clinker increase to each change made to the raw mill. The AFE goal was to increase the raw mill production rate from 130 ton/hr to at least 140 ton/hr, to allow it to keep up with the kiln (Lyons Plant Expansion, Project Scope: Revised February 20, 1998, CEMEX_NEIC_00003449 at 3457; Revised September 21, 1998, CMXLYO00155865 at 155867). As the kiln is limited to 8,064 operating hours (CDPHE, Title V Operating Permit, Last Revised October 8, 2002, US00121584 at 121621), the increase in clinker production facilitated by debottlenecking the raw mill is at least: 10 ton/hr x 8,064 x 0.6 = **48,384 ton/yr**.

5 and 6. The results of my potential increases in NOx emission calculations for individual changes are summarized in Table 8.

Table 8:
Potential Increases in NOx Emissions for Changes Considered Individually
Case B, Step 2

	Project Name	Scenario 1 Increase NOx Emissions (ton/yr) ⁸⁸	Scenario 2 Increase NOx Emissions (ton/yr) ⁸⁹
1	Combustion Chamber Modifications	95	89
2	Raw Mill Transfer Line	*	*
3	Purchased Oxygen Injection	183	169
4	ID Fan 1500 HP Motor	52	52
5	New Raw Feed Compressor	*	*
6	New High Efficiency Separator	212	250
7	Quarry	167	198
8	Tempering Tee Damper ⁹⁰	162*	195*
9	VSA Oxygen Plant	95	112
10	New Annulus Boost Fan	60	56

* Indicates that a change was made to the raw mill. See note 90, below.

Table 7 shows that the increase in NOx emissions from each of the subject changes exceeds the NOx significance threshold of 40 ton/yr, thereby triggering a Step 3 analysis.

As discussed in Section III, minor source permits were obtained for the PM10 emission increases from some of the changes made at the upstream and downstream material handling units. However, CEMEX never permitted the increase in NOx emissions from the pyroprocessing unit, either when modified or when the upstream and downstream units were debottlenecked. Thus, when changes in Table 1 are evaluated individually for PSD applicability, the increase in NOx emissions at these indirect sources must be included in the netting analysis.

⁸⁸ Increase in NOx emissions = [increase in clinker from Table 7 x NOx emission factor from Table 5]/2000 lb/ton.

⁸⁹ Increase in NOx emissions = [increase in clinker from Table 7 x NOx emission factor from Table 6]/2000 lb/ton.

⁹⁰ The changes to the raw mill are #2, #5, and #8. The increase in NOx from these three changes is calculated together (see also Table 7 and note 87) as the record does not allow the increase from each individual change to be estimated. Thus, the increase in NOx emissions is calculated relative to the last change in the series, the tempering tee damper.

2. Netting Analysis for Individual Changes (Case B, Step 3)

Section IV.C.1 demonstrates that all of the changes taken individually would result in a significant increase in NOx emissions, satisfying Step 2. Step 3 requires a determination of the net change in emissions. As explained above in Section IV.B.2, the net emission change equals the emission increase from Step 2, minus source-wide creditable and contemporaneous emission decreases, plus source-wide creditable and contemporaneous emission increases.

I am not aware of any contemporaneous and creditable emission decreases that occurred at the Lyons Plant during the subject period.⁹¹ Purchased oxygen, Change #3, which was in part replaced by the VSA oxygen plant, is not a creditable decrease because it has never been federally enforceable.⁹²

The NOx emission increases for each change are listed in Table 8, in the order of construction. These increases are contemporaneous and creditable increases. Thus, the net increase in emissions is the sum of each change's emissions plus all those that preceded it for pyroprocessing changes. For example, the net increase in NOx emissions for Change #4 is the sum of the NOx emissions from Changes #1, #3, and #4. For Scenario 1, the calculation using emission increases from Table 8 is: $95+183+52 = 330$ ton/yr. For Scenario 2, the calculation is: $89+169+52 = 310$ ton/yr. The results of these Step 3 netting calculations are summarized in Table 9.

⁹¹ See also, Expert Report of Walter L. Greer, p. 33, n. 164.

⁹² NSR Manual, p. A.38.

Table 9:
Net Increase in NOx Emissions for Changes Considered Individually
Case B, Step 3

	Project Name	Increase NOx Scenario 1 (ton/yr)	Increase NOx Scenario 2 (ton/yr)
1	Combustion Chamber Modifications	95	89
2	Raw Mill Transfer Line	95	89
3	Purchased Oxygen Injection	278	258
4	ID Fan 1500 HP Motor	330	310
5	New Raw Feed Compressor	330	310
6	New High Efficiency Separator	330	310
7	Quarry	330	310
8	Tempering Tee Damper	330	310
9	VSA Oxygen Plant	425	422
10	New Annulus Boost Fan	485	478

The total net increases for the last change in Case B, Step 3, the annulus boost fan, range from **478 ton/yr** to **485 ton/yr**. This compares to 465 ton/yr to 594 ton/yr for Case A, Step 3 (Table 4). The differences between these various estimates are immaterial and do not affect my opinion that the collective and individual changes would have resulted in an increase in NOx emissions of a sufficient magnitude to trigger PSD review. These totals fall within the anticipated range for calculations of this type, given the gaps in the record and the variability in NOx emissions. CEMEX's overall production goal for the Lyons Plant does not precisely match the sum of the increases in estimated clinker production from individual changes because all of the changes are not included in my calculations, I used the lower end of the range of production increases for each change, and, based on the available record, it is impossible to estimate precisely the potential increase in clinker from each change.

The net increase in NOx emissions in Table 9 for each individual change exceeds the 40 ton/yr significance threshold. Thus, each change considered alone is a major modification of a major stationary source. Taken individually, PSD would have been triggered for each of these changes. A PSD permit should have been obtained prior to constructing each change, before the dates listed in Table 1.

In sum, regardless of how this collection of changes is viewed, collectively (Case A) or individually (Case B), and irrespective of the baseline (start of construction or standing in shoes of permit engineer) and the production goal (Case A: 546,250 ton/yr – 577,920 ton/yr of clinker), the result is significant net increases of NOx emissions. Thus, the collective changes and each individual change are a major modification of a major stationary source that triggers PSD review. The most appropriate way to address this

collection of changes is as a single project because the changes are interrelated, with the singular purpose of increasing cement production.⁹³

⁹³ Expert Report of Walter L. Greer, p. 26.

Appendix A – Statement of Qualifications

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EDUCATION

Ph.D. Environmental/Civil Engineering, University of California, Berkeley, 1980.
M.S. Environmental/Civil Engineering, University of California, Berkeley, 1975.
B.S. Physics (with high honors), University of Florida, Gainesville, 1971.

REGISTRATION

Registered Professional Engineer: Arizona (Environmental, 2001-present; #36701), California (Chemical, 2002-present; CH 6058), Florida (2001-present; #57886), Georgia (2002-present; #PE027643), Washington (2002-present; #38692), Wisconsin (2005-present; #37595-006)
Board Certified Environmental Engineer, American Academy of Environmental Engineers, Certified in Air Pollution Control (DEE #01-20014), 2002-present
Qualified Environmental Professional (QEP), Institute of Professional Environmental Practice (QEP #02-010007), 2001-present

PROFESSIONAL HISTORY

Consulting Engineer, 1981-present
Lawrence Berkeley National Laboratory, Principal Investigator, 1977-1981
University of California, Berkeley, Program Manager, 1976-1977
Bechtel, Inc., Engineer, 1971-1976, 1964-1966

PROFESSIONAL AFFILIATIONS

American Chemical Society (1981-present)
Phi Beta Kappa (1970-present)
Sigma Pi Sigma (1970-present)

Who's Who Environmental Registry, PH Publishing, Fort Collins, CO, 1992.
Who's Who in the World, Marquis Who's Who, Inc., Chicago, IL, 11th Ed., p. 371, 1993-present.

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National Research Council Committee on Irrigation-Induced Water Quality Problems (Selenium), Subcommittee on Quality Control/Quality Assurance (1985-1990).

National Research Council Committee on Surface Mining and Reclamation, Subcommittee on Oil Shale (1978-80)

REPRESENTATIVE EXPERIENCE

Dr. Fox has over 40 years of experience in the field of environmental engineering, including air pollution control (air permitting; emission calculations; netting analyses; control technology determinations, e.g., BACT, BART, LAER, and MACT; cost-effectiveness analyses; emission offsets; enforceability of air permits; and rulemakings), air quality impacts, ambient monitoring, water quality and water supply investigations, hazardous waste investigations, environmental permitting, nuisance investigations, environmental impact reports, CEQA/NEPA documentation, risk assessments, risk of upset analyses, and litigation support. This work has been performed for numerous private sector, public sector, and public interest clients.

Dr. Fox has performed these types of investigations for a wide range of industrial and commercial facilities including: petroleum refineries and upgrades thereto; reformulated fuels projects; petroleum distribution terminals; shale oil plants; coal gasification & liquefaction plants; conventional and thermally enhanced oil production; underground storage tanks; pipelines; gasoline stations; landfills; railyards; hazardous waste treatment facilities; nuclear, hydroelectric, geothermal, wood, waste, gas, oil, coke and coal-fired power plants; transmission lines; airports; hydrogen plants; petroleum coke calcining plants; coke plants; activated carbon manufacturing facilities; asphalt plants; cement plants; incinerators; flares; manufacturing facilities (e.g., semiconductors, electronic assembly, aerospace components, printed circuit boards, amusement park rides); lanthanide processing plants; ammonia plants; nitric acid plants; urea plants; food processing plants; almond hulling facilities; composting facilities; grain processing facilities; grain elevators; ethanol production facilities; soy bean oil extraction plants; biodiesel plants; paint formulation plants; wastewater treatment plants; marine terminals and ports; gas processing plants; steel mills; iron nugget production facilities; pig iron plant based on blast furnace technology; direct reduced iron plant; acid regeneration facilities; railcar refinishing facility; battery manufacturing plants; pesticide manufacturing and repackaging facilities; pulp and paper mills; selective catalytic reduction (SCR) systems; halogen acid furnaces; contaminated property redevelopment projects (e.g., Mission Bay, Southern Pacific Railyards,

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Moscone Center expansion, San Diego Padres Ballpark); residential developments; commercial office parks, campuses, and shopping centers; server farms; transportation plans; and a wide range of mines including sand and gravel, hard rock, limestone, nacholite, coal, molybdenum, gold, zinc, and oil shale.

PUBLICATIONS AND PRESENTATIONS (Partial List - Representative Publications)

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Deposition of David Repasz, September 15, 2010

Deposition of Ram Seetharam, August 11 and 12, 2010

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Deposition of Randal Wiley, November 10, 2010

Deposition of Gregory Wright, November 9, 2010

Appendix C – Expert Testimony

Cases in which I have provided testimony at trial and in administrative hearings between February 2007 and February 2011 include the following:

Friends of the Chatahooche and Sierra Club v. Dr. Carol Couch, Director, Environmental Protection Division of Natural Resources Department, Respondent, and Longleaf Energy Associates, Intervener, September - October 2007.

In the Matter of Application for a Certificate of Authority to Install Wet Flue Gas Desulfurization and Selective Catalytic Reduction Facilities and Associated Equipment for Control of Sulfur Dioxide and Nitrogen Oxide Emissions at Oak Creek Power Plant Unit 5, 6, 7, and 8, Docket No. 6630-CE-299, February 2007.

United States et al. v. Cinergy et al., Wabash remedy phase, Southern District of Indiana, Indianapolis Division, Civil Action No. IP99-1693 C-M/S, July 2008 - February 2009.

In Re: Southwestern Electric Power Company, Arkansas Pollution Control and Ecology Commission, Consolidated Docket No. 08-006-P, June 2007 - June 2009.

A. Singh v. S. Assaedi, Contra Costa County Superior Court, CA, July 2009.

In the Matter of Application for Las Brisas Energy Center, LLC for State Air Quality Permit 85013 and Prevention of Significant Deterioration Air Quality Permit PSD-TX-1138; for Hazardous Air Pollutant Major Source Permit HAP48 and Plantwide Applicability Limit PAL41, before the Texas State Office of Administrative Hearings, August - November 2009.

Sierra Club, Ohio Valley Environmental Coalition, Coal River Mountain Watch, West Virginia Highlands Conservancy v. John Benedict, Director, Division of Air Quality, West Virginia Department of Environmental Protection and TransGas Development Systems, LLC, Appeal No. 10-01-AQB. October 2010.

Cases in which I have provided depositions, affidavits, and expert reports between February 2007 and February 2011 include:

Expert Report in April 2006, Rebuttal Report in January 2007, and Deposition in March 2007 on behalf of Plaintiffs in United States, New York et. al., Ohio Citizen Action et al., v. American Electric Power Service Corp., U.S. District Court for the Southern District of Ohio, Eastern Division, Consolidated Civil Action Nos. C2-99-1182 and C2-99-1250.

Expert Report and Deposition in February 2007 on behalf of Plaintiffs in Re PSD Construction Permit Issued to Great Plains Energy, Kansas City Power & Light – Iatan Generating Station, Sierra Club v. Missouri Department of Natural Resources, Great Plains Energy, and Kansas City Power & Light.

Deposition on March 30, 2007 on behalf of Petitioners in the Matter of Objection to the Issuance of New Construction Federally Enforceable State Operating Permit No. 075-22858-00032, United Association of Plumbers and Pipefitters, Local Union 661 et al. v. Premier Ethanol and Indiana Department of Environmental Management.

Expert Disclosure in September 2007 on behalf of Plaintiffs in Friends of the Chatahooche and Sierra Club v. Dr. Carol Couch, Director, Environmental Protection Division of Natural Resources Department, Respondent, and Longleaf Energy Associates, Intervener.

Expert Report in December 2007 on behalf of Intervenors In the Matter of Application for a Certificate of Authority to Install Wet Flue Gas Desulfurization and Selective Catalytic Reduction Facilities and Associated Equipment for Control of Sulfur Dioxide and Nitrogen Oxide Emissions at Oak Creek Power Plant Unit 5, 6, 7, and 8, Docket No. 6630-CE-299.

Expert Report in July 2007 on behalf of Plaintiffs in United States v. CEMEX California Cement, LLC CV 07-223 (C.D. Cal).

Expert Report in August 2008, Rebuttal Report in October 2008, Deposition in October 2008, Testified in February 2009 on behalf of Plaintiffs in Wabash River remedy phase of U.S. et al. v. Cinergy et al., In Southern District of Indiana, Indianapolis Division, Civil Action No. IP99-1693 C-M/S.

Prefiled direct testimony in March 2009, prefiled rebuttal testimony in May 2009, testified June 2009 on behalf of Plaintiffs in Southwestern Electric Power Company.

Expert Report in January 2009 , Rebuttal Report in May 2009, Deposed May 2009 on behalf of Petitioners. In the Matter of Objection to the Issuance of Significant Source Modification Permit No. 089-25484-00453 to BP Products North America Inc., Whiting Business Unit, Save the Dunes Council, Inc., Sierra Club., Inc., Hoosier Environmental Council et al., Petitioners, B. P. Products North American, Respondents/Permittee, before the Indiana Office of Environmental Adjudication.

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Electric Co., U.S. District Court, Northern District of California (Case No. C-09-4503 SI.)

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Expert Report November 15, 2010 and affidavit November 24, 2010 in Sierra Club v. City of Holland and Holland Board of Public Works, Case No. 1:08-cv-1183, U.S. District Court for the Western District of Michigan.

Appendix D – Exhibits

1. Baseline and Potential to Emit (PTE) Emissions Spreadsheet
2. Statistical Analyses of Variance

Scenario 1. Case B: As of the Date of Construction

	Annual CMXLYO001 (ton/mo)	Annual Clinker (ton/yr)	Monthly NOx (lb/mo)	Annual NOx (ton/mo)	Baseline Clinker (ton/yr)	Baseline NOx (ton/yr)	Baseline Emission Factor (lb/ton)	Production Increase (ton/yr)	Increase in NOx (ton/yr)	Modification
Jan-94	41,000		166.1				US00000583			
Feb-94	34,000		137.8				US00000583			
Mar-94	39,000		158.0				US00000583			
Apr-94	35,000		141.8				US00000583			
May-94	39,000		158.0				US00000583			
Jun-94	33,000		133.7				US00000583			
Jul-94	14,000		56.7				US00000583			
Aug-94	34,000		137.8				US00000583			
Sep-94	36,000		145.9				US00000583			
Oct-94	39,000		158.0				US00000583			
Nov-94	38,000		154.0				US00000583			
Dec-94	37,000	418,505	149.9	1,695.8			US00000583, US00006694			
Jan-95	37,054	419,000	132.0	1,697.8						
Feb-95	30,620		109.1							
Mar-95	39,669		141.3							
Apr-95	38,131		135.8							
May-95	22,002		62.0							
Jun-95	36,378		88.1							
Jul-95	36,493		137.1							
Aug-95	38,358		122.5							
Sep-95	36,220		90.0							
Oct-95	32,793		92.3				CMXLYO00179838, LYO-19			
Nov-95	39,040		199.7				CMXLYO00179839			
Dec-95	38,516	425,274	193.0	1,514.7						
Jan-96	37,804		168.4							
Feb-96	35,973		164.3							
Mar-96	8,591		9.8							
Apr-96	38,188		118.4							
May-96	38,374		175.0							
Jun-96	37,466		166.0							
Jul-96	40,607		182.6							
Aug-96	39,906		173.1							
Sep-96	38,750		177.9							
Oct-96	38,090		175.6							
Nov-96	38,003		200.0							
Dec-96	38,556	430,308	161.4	1,872.5						
Jan-97	38,468		182.3							
Feb-97	35,105		178.2							
Mar-97	26,639		97.4		430,741	1,747	8.11	23,386	95	Combustion Chamber Modification
Apr-97	25,268		73.8							
May-97	42,324		141.5		417,794	1,694	8.11	45,158	183	Routine Oxygen Injection & Raw Mill Transfer Line (same start of construction date)
Jun-97	47,772		126.5					48,384		Raw Mill Tranfer Line
Jul-97	36,419		108.3							
Aug-97	41,484		151.2		433,615	1,739	8.02	12,902	52	ID Fan 1500 HP Motor Install
Sep-97	41,224		160.2							
Oct-97	43,834		156.0							
Nov-97	35,157		147.1							
Dec-97	45,000	458,694	178.9	1,701.4	441,259	1,794	8.13	48,384		New Raw Feed Compressor
Jan-98	41,801		140.1							
Feb-98	42,938		136.2							
Mar-98	46,277		167.8							
Apr-98	26,572		105.8							
May-98	31,392		94.6							
Jun-98	40,906		123.5							
Jul-98	46,477		94.3							
Aug-98	44,300		50.1							
Sep-98	47,000		102.5							
Oct-98	44,976		193.4		470,503	1,627	6.91	61,286	212	New High Efficiency Separator & Quarry
Nov-98	42,983		161.8					48,384	167	Quarry
Dec-98	45,386	501,008	137.2	1,507.3						
Jan-99	24,535		98.6		479,851	1,604	6.69	48,384	162	Tempering Tee Damper
Feb-99	41,116		172.3							
Mar-99	43,014		203.5							
Apr-99	38,429		159.4							
May-99	44,070		211.7		490,658	1,655	6.75	28,224	95	VSA Oxygen Plant
Jun-99	40,786		104.0							
Jul-99	41,901		148.4							
Aug-99	46,262		199.9							
Sep-99	41,960		154.2							
Oct-99	45,064		176.1							
Nov-99	43,134		169.5							
Dec-99	49,760	500,031	176.1	1,973.7						
Jan-00	24,120		92.7							
Feb-00	42,791		182.3							
Mar-00	49,511		203.5							
Apr-00	48,115		159.4							
May-00	47,335		191.8							
Jun-00	44,287		159.4							
Jul-00	48,384		203.5							
Aug-00	45,183		187.7		514,610	1,906	7.41	16,128	60	New Annulus Boost Fan
Sep-00	43,860		190.0							
Oct-00	51,521		239.5							
Nov-00	44,883		181.9							
Dec-00	50,002	539,992	170.6	2,162.3						

- Substitutions for missing data
- Debottlenecking Chanages
- Raw Mill Debottlenecking Changes
- Pyroprocessing Changes
- Substitutions used other cites from record.

Clinker from Clinker Production - Historical Summary, CMXLYO00628849, unless otherwise noted.
NOx emissions from NOx Emissions - Kiln - Historical Summary, CMXLYO00628845, unless otherwise noted.

Scenario 2: Case B: Standing in the Shoes of the Permit Engineer

	Clinker (ton/mo)	Annual Clinker (ton/yr)	Monthly NOx (lb/mo)	Annual NOx (ton/mo)	Baseline Clinker (ton/yr)	Baseline NOx (ton/yr)	Baseline Emission Factor (lb/ton)	Production Increase (ton/yr)	Increase in NOx (ton/yr)	Modification
Jan-94	41,000		166.1				US00000583			
Feb-94	34,000		137.8				US00000583			
Mar-94	39,000		158.0				US00000583			
Apr-94	35,000		141.8				US00000583			
May-94	39,000		158.0				US00000583			
Jun-94	33,000		133.7				US00000583			
Jul-94	14,000		56.7				US00000583			
Aug-94	34,000		137.8				US00000583			
Sep-94	36,000		145.9				US00000583			
Oct-94	39,000		158.0				US00000583			
Nov-94	38,000		154.0				US00000583			
Dec-94	37,000	418,505	149.9	1,695.8			US00000583, US00006694			
Jan-95	37,054	419,000	132.0	1,697.8						
Feb-95	30,620		109.1							
Mar-95	39,669		141.3							
Apr-95	38,131		135.8							
May-95	22,002		62.0							
Jun-95	36,378		88.1							
Jul-95	36,493		137.1							
Aug-95	38,358		122.5							
Sep-95	36,220		90.0							
Oct-95	32,793		92.3				CMXLYO00179838, LYO-19			
Nov-95	39,040		199.7				CMXLYO00179839			
Dec-95	38,516	425,274	193.0	1,514.7						
Jan-96	37,804		168.4							
Feb-96	35,973		164.3							
Mar-96	8,591		9.8							
Apr-96	38,188		118.4							
May-96	38,374		175.0							
Jun-96	37,466		166.0							
Jul-96	40,607		182.6							
Aug-96	39,906		173.1							
Sep-96	38,750		177.9							
Oct-96	38,090		175.6							
Nov-96	38,003		200.0							
Dec-96	38,556	430,308	161.4	1,872.5						
Jan-97	38,468		182.3							
Feb-97	35,105		178.2							
Mar-97	26,639		97.4		421,890	1,605	7.61	23,386	89	Combustion Chamber Modification
Apr-97	25,268		73.8							
May-97	42,324		141.5		407,915	1,529	7.50	45,158	169	Routine Oxygen Injection & Raw Mill Transfer Line (same start of construction date)
Jun-97	47,772		126.5					48,384		Raw Mill Transfer Line
Jul-97	36,419		108.3							
Aug-97	41,484		151.2		433,615	1,739	8.02	12,902	52	ID Fan 1500 HP Motor
Sep-97	41,224		160.2							
Oct-97	43,834		156.0							
Nov-97	35,157		147.1							
Dec-97	45,000	458,694	178.9	1,701.4	427,013	1,682	7.88	48,384		New Raw Feed Compressor
Jan-98	41,801		140.1							
Feb-98	42,938		136.2							
Mar-98	46,277		167.8							
Apr-98	26,572		105.8							
May-98	31,392		94.6							
Jun-98	40,906		123.5							
Jul-98	46,477		94.3							
Aug-98	44,300		50.1							
Sep-98	47,000		102.5							
Oct-98	44,976		193.4		437,680	1,788	8.17	61,286	250	New High Efficiency Separator & Quarry Quarry
Nov-98	42,983		161.8					48,384	198	
Dec-98	45,386	501,008	137.2	1,507.3						
Jan-99	24,535		98.6		444,501	1,787	8.04	48,384	195	Tempering Tee Damper
Feb-99	41,116		172.3							
Mar-99	43,014		203.5							
Apr-99	38,429		159.4							
May-99	44,070		211.7		463,017	1,831	7.91	28,224	112	VSA Oxygen Plant
Jun-99	40,786		104.0							
Jul-99	41,901		148.4							
Aug-99	46,262		199.9							
Sep-99	41,960		154.2							
Oct-99	45,064		176.1							
Nov-99	43,134		169.5							
Dec-99	49,760	500,031	176.1	1,973.7						
Jan-00	24,120		92.7							
Feb-00	42,791		182.3							
Mar-00	49,511		203.5							
Apr-00	48,115		159.4							
May-00	47,335		191.8							
Jun-00	44,287		159.4							
Jul-00	48,384		203.5							
Aug-00	45,183		187.7		490,779	1,699	6.92	16,128	56	New Annulus Boost Fan
Sep-00	43,860		190.0							
Oct-00	51,521		239.5							
Nov-00	44,883		181.9							
Dec-00	50,002	539,992	170.6	2,162.3						

- Substitutions for missing data
- Debottlenecking Chanages
- Raw Mill Debottlenecking Changes
- Pyroprocessing Changes
- Alternatives for filled values

Exhibit 2: Statistical Analyses

DATA:

Year	Clinker (ton/yr)	NOx (ton/yr)	NOx (lb/ton)
1994	418,505	1696.0	8.1
1995	425,274	1514.7	7.1
1996	430,272	1872.5	8.7
1997	458,694	1701.4	7.4
1998	501,008	1507.3	6.0
1999	500,031	1973.7	7.9
2000	539,992	2162.2	8.0
2001	492,855	2362.0	9.6
2002	472,945	1747.1	7.4
2003	516,251	1835.0	7.1
2004	446,097	1708.9	7.7
2005	440,383	1591.3	7.2
2006	466,173	2011.7	8.6
2007	479,224	1688.6	7.0
2008	478,520	1295.5	5.4

NOTES:

Clinker production for 1995 - 2008 from: Estimated Clinker Production - Historical Summary, CMXLYO00628849. Clinker production for 1994 from US00006692 at 6694;

NOx Emissions for 1995-2008 from: NOx Emissions - Kiln - Historical Summary, CMXLYO00628845. NOx emissions for 1994 is from US00006692 at 6694.

NOx emissions without moisture correction for 2000 - 2001 based on: Kendrick Deposition, p. 34:12-23; Memorandum from Matthew Parks to Paul Carr Re: CEMEX CEMS, September 10, 2007, CEMEX_NEIC_00008623 at 8624-8626.

Use of CEMEX's historical summaries for clinker production and NOx emissions is done for convenience and should not be construed as an endorsement of those estimates.

NOx emissions expressed in lb/ton is calculated from: (NOx emissions ton/yr)/(2000 lb/ton)/clinker production ton/yr).

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.10138212
R Square	0.01027833
Adjusted R Square	-0.0658541
Standard Error	1.06965864
Observations	15

ANOVA

	df	SS	MS	F	Signific. F
Regression	1	0.15447	0.15447	0.135006	0.7192095
Residual	13	14.87421	1.14417		
Total	14	15.02867			

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	8.96137676	3.830658	2.339383	0.03593	0.6857434	17.23701	0.6857434	17.23701
Clinker (ton/yr)	-2.98E-06	8.11E-06	-0.36743	0.719209	-2.05E-05	1.45E-05	-2.05E-05	1.454E-05

RESIDUAL OUTPUT

Observation	Predicted NOx (lb/ton)	Residuals	Standard Residuals
1	7.71	0.39	0.38
2	7.69	-0.57	-0.55
3	7.68	1.02	0.99
4	7.59	-0.18	-0.17
5	7.47	-1.45	-1.41
6	7.47	0.42	0.41
7	7.35	0.68	0.66
8	7.49	2.10	2.04
9	7.55	-0.16	-0.16
10	7.42	-0.31	-0.30
11	7.63	0.03	0.03
12	7.65	-0.42	-0.41
13	7.57	1.06	1.03
14	7.53	-0.49	-0.47
15	7.54	-2.12	-2.06

PROBABILITY OUTPUT

Percentile	NOx (lb/ton)
3.33	5.41
10.00	6.02
16.67	7.05
23.33	7.11
30.00	7.12
36.67	7.23
43.33	7.39
50.00	7.42
56.67	7.66
63.33	7.89
70.00	8.03
76.67	8.11
83.33	8.63
90.00	8.70
96.67	9.59